	= 0.25	
2a	Product of the force and the perpendicular distance from the point.	A1
2bi	Taking moments about Q (moment due to weight =) 1.2 × 270 × 9.81 (moment due to post =) 1800 × (1.6 / cos θ) 1.2 × 270 × 9.81 = 1800 × (1.6 / cos θ) θ = 25° <u>*Alternative solution*</u> moment due to weight = $(270 \times 9.81 \cos\theta)(\frac{1.2}{\cos\theta})$ moment due to post = $1800(\frac{1.6}{\cos\theta})$ OR moment due to weight = $(270 \times 9.81 \cos\theta)(1.2 \cos\theta)$ moment due to post = $1800 \times 1.6 \cos\theta$	C1 C1 A1
2bii	Vertical component of F = 1800 cos 25° = 1631 N Weight = 270 x 9.81 = 2649 N For vertical equilibrium (or net force zero in vertical direction), R must have a vertical component R is not horizontal OR Taking moments about CG, there is anticlockwise moment due to F, so R must provide clockwise moment R is not horizontal OR By drawing the lines of action of the three concurrent forces they must meet at a point (must indicate clearly on diagram) R is not horizontal	M1 A1 (M1) (A1) (M1) (A1)